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Evolution of the gas properties of two simulated colliding galaxies

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ABSTRACT

The collision of galaxies, or mergers, is one of the main processes that drive the formation and evolution of galaxies. The formation of new stars depends strongly on the characteristics of the gas present in a galaxy. But, due to the variety of possible collision scenarios, is still not fully understood how mergers affect the gas physical properties. In this work, we want to understand how the gas properties of a galaxy are affected during a merger interaction. For this purpose, we are using a non-cosmological simulation of two colliding galaxies that are simulated with the GADGET-4 code. They present supernova feedback and radiative cooling mechanisms to reproduce star formation effects. In this simulation we determined that the gas is affected by the interaction of both galaxies, we identify the key moments of star formation and estimate relevant physical properties in this stages. After the galaxies approach, an starburst event is triggered, probably due to the compression of cold gas clouds. But then, the gas transitioned to a quenching phase. This suggest that the merger and the SN feedback perturbed and heat the gas after the interaction.

1. Introduction.

Galaxy mergers and interactions are fundamental processes of galactic formation and evolution. According to Pearson et al. (2019), in the framework of cold dark matter cosmology, the merging of dark matter halos occurs through hierarchical growth, leading to the convergence of their baryonic counterparts (Conselice 2014; Somerville & Davé 2015). This merging process entails the disruption of central galaxies within the dark matter halos due to the exertion of tidal forces, which result in the displacement of stars from the disk to the spheroidal component (Toomre & Toomre 1972).

Mergers are also believed to instigate episodes of intense star formation, known as starbursts. Simulations indicate that these starbursts arise from the tidal interactions of merging galaxies, which compress and shock the gas, leading to the rapid generation of stars (e.g. Barnes 2004). This association between starbursts and merging galaxies has given rise to the prevailing hypothesis that the majority of merging galaxies undergo a starburst phase (Joseph & Wright 1985; Schweizer 2005). However, other studies suggest that after this starburst events, galaxies transition to a quenching phase, due to the formation of gravitational potentials that prevents gas to collapse (Petersson et al. 2023).

Early investigations utilizing N-body simulations of idealized mergers demonstrated the profound distortion and deformation of merger remnants by tidal forces (Toomre & Toomre 1972). Both theoretical studies (e.g. Moreno et al. 2015) and observational evidence (Patton et al. 2016) indicate that the final morphological outcome of the interaction is influenced by the orbital properties and mass ratio of the progenitors, as well as the physical characteristics of the involved galaxies. Due to this important dependence on the orbital parameters and mass ratios, there is a huge diversity on galaxies whose physical properties are still not fully understood. Understanding the physical aspects that regulate or increase star formation in galaxies is one of the most critical aspects of understanding galaxy evolution.

During the last few decades the state-of-the-art of numerical simulations, together with the improvements in the computational power, evolved into more efficient and accurate codes. They can reproduce different properties, such as the dynamics of dark matter and the baryonic physics, that we see in the universe (Vogelsberger et al. 2020, and references therein). GADGET is a code widely used to simulate the formation of the universe in different contexts, including cosmological ones. In this work we are gonna be using the last release of this code, GADGET-4 (Springel et al. 2021), in order to understand the effects of merger interactions on star formation and different gas properties.

2. GADGET-4.

We are using a non-cosmological N-body simulation of two galaxies implemented with GADGET-4. This Lagrangian code implements a TreePM approach for computing gravitational forces and a smoothed particle hydrodynamics(SPH) treatment for the gas dynamics, more details on this implementation can be found in Springel et al. (2021). The simulation that we are working with include a simple treatment for star formation and supernovae feedback (Springel & Hernquist 2003), together with a radiative cooling model due to helium and hydrogen (Katz et al. 1996). In this sense the ISM is treated in a multiphase perspective, allowing star formation in highly dense cold clouds. At the same time, this regions can be heated by supernovae self-regulating star formation.

2.1. Colliding Galaxies

The colliding galaxies implementation used in this work, is part of the examples that GADGET-4 provided to test the code. The processes of cooling and star formation are included in order to test the global properties of the gas during a merger interaction. This two galaxies will evolved in 38 snapshots, that represent $\sim$ 10 Gyrs of evolution. In Figure 1 we can see the different particles of the simulation, the top panels show the dark matter halo that defines the initial structures, and also the gas that will formed(or not) stars. Notice that the gas start in a very well-defined disk, that will lead to high star formation rates(SFR) at the beginning of the simulation. In the bottom panels we can

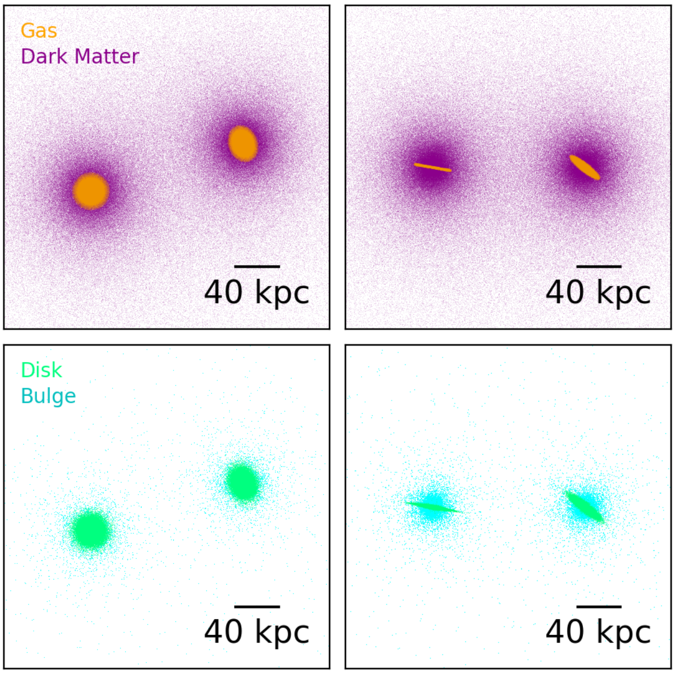

Fig. 1. Initial particle distributions, before starting star formation. Top panels: face-on and edge-on distributions of gas(orange) and dark matter (purple) particles. In this well settled disk of gas, stars will start to form. Lower panels: face-on and edge-on distributions of disk (limegreen) and bulge (cyan) stellar particles, this stellar populations will affect the gravitational potentials, i.e the interaction between the galaxies.

also see the initial distributions of stellar particles, from which the simulation will evolve.

3. Results

3.1. SFR and key moments

To identify the key moments of evolution of the galaxy, in Figure 2 we are showing the SFRs and the morphologies of the two galaxies at different stages. Each moment was chosen according to the local maximum and minimum of the SFR

At the beginning of the simulation, galaxies are relatively isolated, each exhibiting high levels of star formation with an SFR of about 35 $M_{\odot}yr^{-1}$. This high SFR indicates a vigorous star formation within each galaxy fueled by the presence of big gas reservoirs, that are given by the initial setup of the simulation. The second key moment shows the first local minimum in the SFR. As both galaxies approach each other and begin their first passage, the interaction between them triggers complex gravitational forces that result in gas disruption which in addition to supernova feedback, generate a lower SFR of 10 $M_{\odot}yr^{-1}$. Supernova feedback expels material and energy into the ISM, creating regions of hot gas that can disrupt the collapse of gas clouds and inhibit the formation of new stars. The third important moment of the interaction occurs after the first passage, when both galaxies are momentarily separated before beginning to strongly interact again. This phase sees a peak in the SFR, reaching approximately 20 $M_{\odot}yr^{-1}$, as gas can cool and compress again, leading to a burst in star formation. The fourth key moment occurs after the second passage, where the SFR become more or less constant at ~ 5 $M_{\odot}yr^{-1}$. This decrease is likely due to the depletion of cold gas reservoirs and the disruption caused by the ongoing merger process. Even as the merger progresses and the galaxies become more entwined, there is a final surge in star formation activity. The fifth key moment corresponds to the last notable peak, reaching approximately 10 $M_{\odot}yr^{-1}$. This surge can be attributed to the interaction of gas rich regions. As the galaxies continue to merge and their central regions coalesce, dense gas clouds located in the central regions of the merging galaxies are likely to merge as well. Processes that might also contribute are the formation of tidal tails, that result in dense regions of gas. The final key moment corresponds to the last snapshot, with a SFR of less than 1 $M_{\odot}yr^{-1}$. This decline is probably caused by the consumption of the most part of available cold gas and the heating due to SN feedback, quenching star formation activity. Also it is possible that the merger-induced shocks and disruptions may have redistributed the gas in a less favorable manner for star formation.

3.1.1. SFR temperature vs density

As we previously discuss, the gas distribution and star formation change as a function of time evolution and the interaction of both galaxies. This can be related to the change in density and temperature of the gas, that we can see in Figure 3. This figure show us both properties as a function of the snapshot of the simulation, color-coded by the particles in each bin.

At the beginning of the simulation, clouds are highly dense and have lower temperatures, this due to the initial properties that were given. At the snapshot 9 an small fraction of particles moved to higher temperatures, as we can see from the bluer color of this region. This can be explained due to the high star formation and the SN feedback from the new stars, that heat the gas. This snapshot is also close to the first pericenter of the two galaxies. After this passage, in snapshot 12, some of the gas particles moved to lower temperatures and between T $\sim$ [-8,-7] K the high density part of the plot is slightly more populated. In this snapshot we can also see an increase in SFR (Figure 2). The star formation event at this point is likely to happen due to the interaction of both galaxies, because of the dense clouds that are generated due to processes such as inflows and turbulence. The star formation event does not happen immediately because time is needed for the cooling of the gas clouds.

The second passage of the galaxies happen in snapshot 25, but in this point there is a much more relevant fraction of stars with higher temperatures. This probably caused by the increase in the SFR and the SN feedback. Another important effect that can explain this temperatures in this moment of evolution, is the gas disruption due to the gravitational interaction of both galaxies. After the passage, in snapshot 28, an small arise of star formation occur. If we compare the properties of the gas of this snapshot with the previous star formation event (snapshot 12), now we have much lower reservoir of cold gas to form stars. Thus, leading to a much smaller star formation event. Finally in the last snapshot, after the effects of tidal forces and SN feedback, both galaxies ends up with a very high fraction of heated gas clouds with low density, explaining why the star formation is also very low at this point.

4. Conclusions

Galaxy mergers are an important part of how galaxies grow and evolve over the history of the universe. It is known that mergers do indeed influence SFR in merging galaxies (Pearson

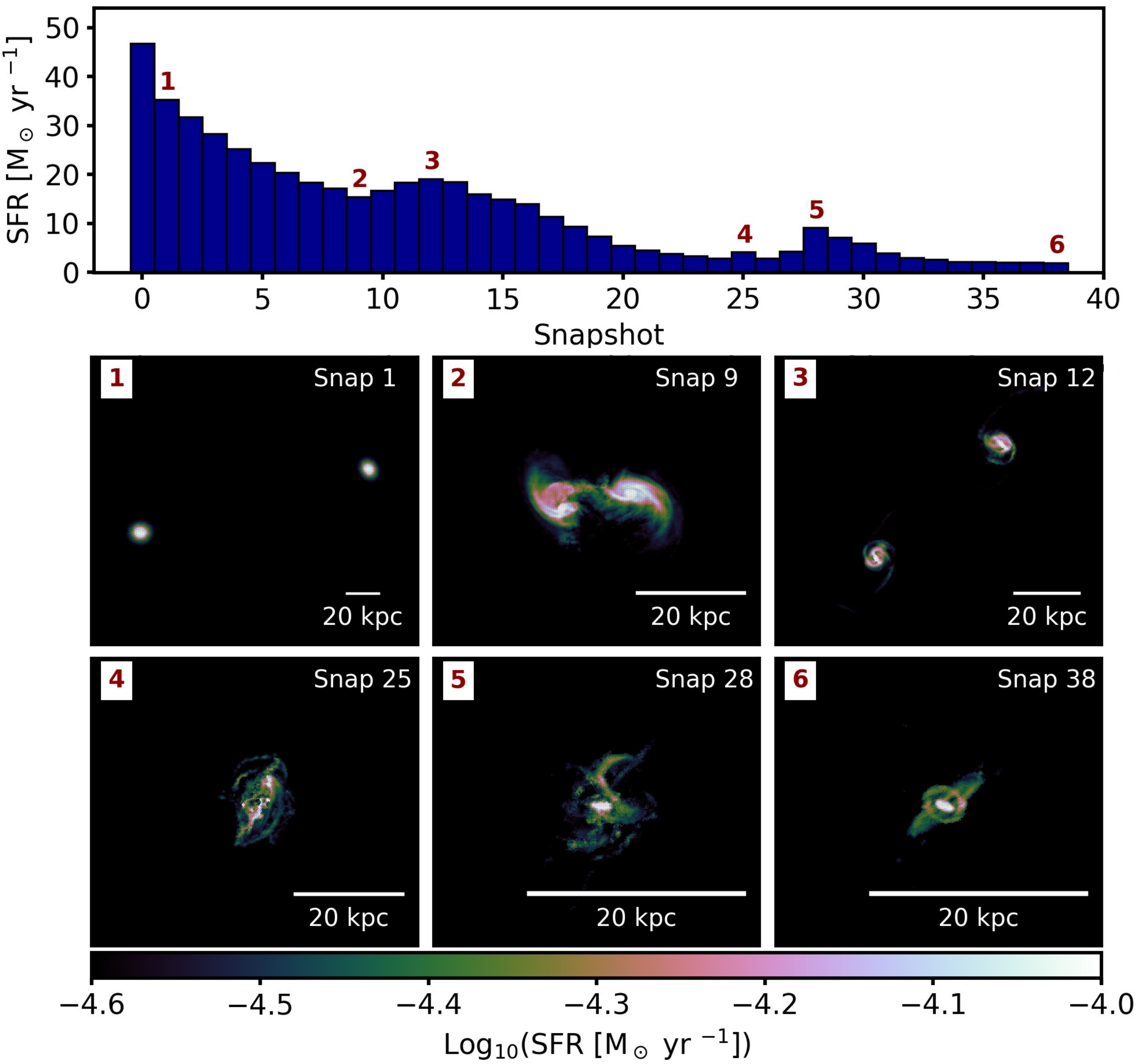

Fig. 2. Upper panel: global SFR of the gas for both galaxies at each snapshot. Relevant moments of formation are highlighted in the snapshots 1, 9, 12, 25, 28 and 38. Lower panels: For each highlighted moment is show the morphologies of the galaxies with hexagonal binning, color-coded by the median SFR of the gas particles in each bin.

et al. 2019). This influence, expressed as a higher SFR (Rodríguez Montero et al. 2019), tends to fight the downward trend of the SFR with occasional starbursts. In this work we analyzed a particular scenario of two merging galaxies, identifying key moments of star formation and the change in different gas properties due to the interaction.

i) During the merger process different important moments of star formation can be identify. As both galaxies get closer, the gravitational interactions lead to the compression of the cold gas clouds, inducing starburst events. The properties of this events will depend on the density and temperature that the gas have previous to the passage.
ii) After the merger, the gas is redistributed and due to the star formations events the gas is consumed. As gas reservoirs are depleted through the formation of stars and the merger process itself, the available fuel for star formation diminishes.
iii) During star formation, massive stars undergo supernova explosions, releasing energy and ejecting material into the surrounding ISM. This feedback heats and disperse gas, inhibiting its collapse. It is important to note that these feedback mechanisms become more pronounced due to the increased star formation activity, leading to the aforementioned SFR decrease over time.

This simulation implement a simple treatment of star formation, SN feedback and radiative cooling, but it shows how the different gas properties can be affected during a merger. The gas spatial distributions, density and temperature change during an interaction, and this changes are related either to an increase

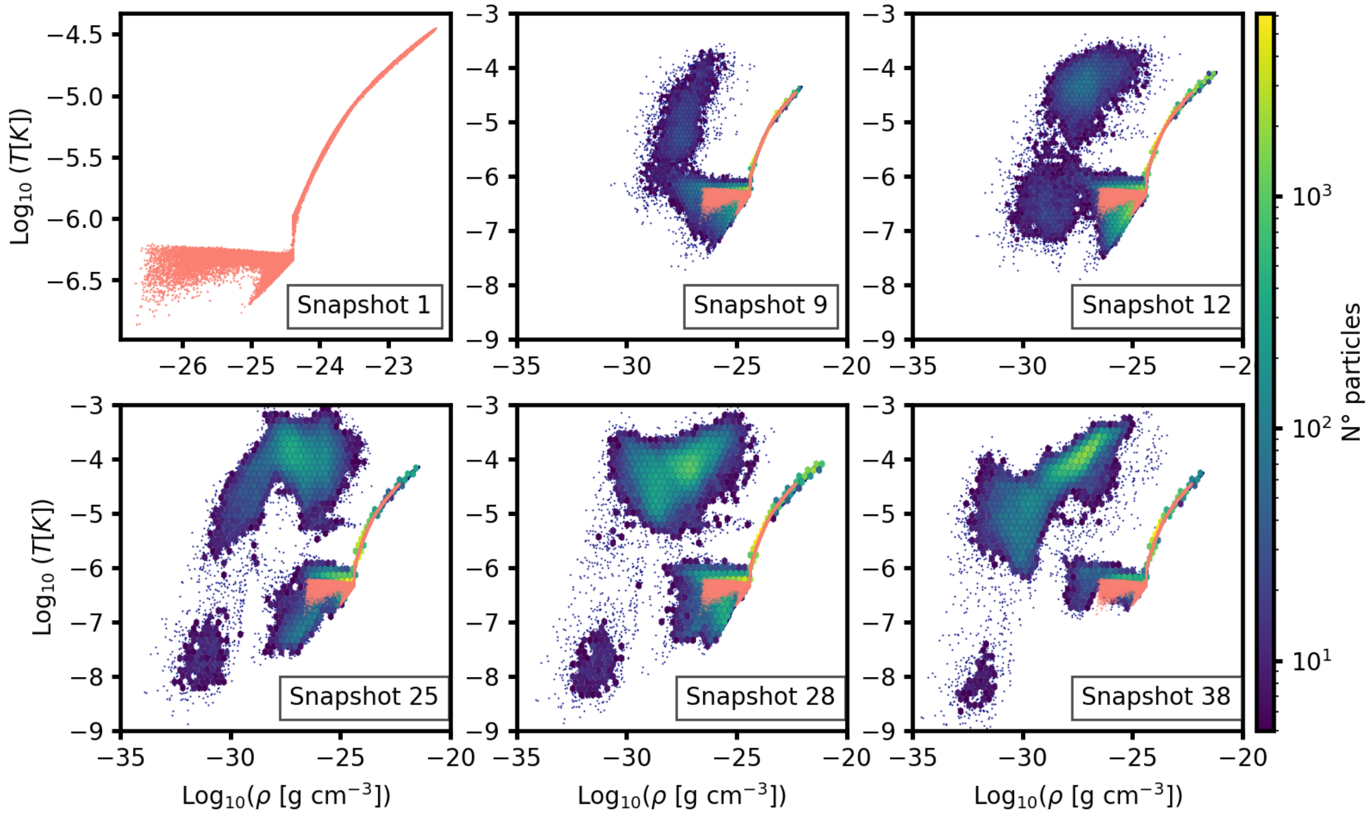

Fig. 3. Temperature of the gas particles as a function of it density for different relevant moments of star formation. This color-coded by the number of particles in each hexagonal bin. To compare the evolution of this gas properties, the snapshot 1 is shown in all plots.

or a decrease in star formation. Is important to highlight that in this work we are presenting a particular case of a merger, with well-defined properties and in a non-cosmological context. In order to understand how different mergers affects the properties of galaxies, a cosmological sample that reproduces different merger ratios and orbital parameters is needed. Moreover GADGET-4 have shown to be useful on the development of simulations that reproduces this complex scenarios in agreement to previous studies (eg. Barnes 2004; Petersson et al. 2023).

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